Sparse, Low-Rank, and Deep: A Unified Compressed-Sensing Framework for Music Source Separation

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Abstract

Music source separation, the recovery of individual stems such as vocals and accompaniment from a single mixture, is usually approached either by classical signal-processing decompositions or by large end-to-end neural networks. We show that a wide span of these methods are instances of one underlying principle: recovery of structured (sparse or low-rank) representations of the time-frequency spectrogram. Under this view we construct and evaluate four separators on a common short-time Fourier transform front end: a low-rank-plus-sparse decomposition by principal component pursuit, a supervised non-negative matrix factorization with per-source dictionaries, a discriminative K-SVD dictionary model with sparse-representation classification, and a deep sparse coding network that predicts a time-frequency mask by unrolling a non-negative proximal-gradient solver and learning its dictionaries end-to-end. We further give a joint-sparsity extension for stereo recovery. On 25 held-out tracks of the MUSDB18 benchmark, the methods form a clear ordering: training-free baselines (median-filtering harmonic percussive separation, REPET-SIM) trail supervised dictionary methods, and the unrolled deep sparse coding network is the strongest non-oracle separator on both sources (vocal SI-SDR 1.6 dB, accompaniment SI-SDR 7.7 dB) while being the fastest learned model at inference. We connect each method to the recovery theory that governs it and report a negative result: the learned dictionaries are highly coherent ($\mu \approx 0.85$), so worst-case exact-recovery guarantees are vacuous at the operating sparsity, yet separation succeeds, an instance of the average-case-versus-worst-case gap for trained dictionaries. Code, trained models, and audio are released.

1. Introduction

Single-channel music source separation asks for the constituent stems of a recording given only the final mixture. It underpins remixing, upmixing, music education, and downstream music-information-retrieval tasks such as transcription and lyric alignment, and it remains hard because the mixing process is information-destroying: many stem configurations explain the same mixture (Cano et al., 2019; Araki et al., 2025). Progress has come from two largely separate traditions. The first is model-based: non-negative matrix factorization (NMF) (Lee & Seung, 1999; Smaragdis, 2007), low-rank-plus-sparse decompositions (Huang et al., 2012), and dictionary learning (Aharon et al., 2006) exploit explicit structural priors on the spectrogram. The second is data-driven: deep networks trained end-to-end, including U-Nets (Jansson et al., 2017; Stoller et al., 2018), recurrent masking models (Stöter et al., 2019), and hybrid waveform/spectrogram architectures (Défossez et al., 2021; Rouard et al., 2023), define the current state of the art.

This paper takes the position that these traditions are two ends of a single axis. The spectrogram magnitude of a musical source is approximately sparse in a source-adapted basis (a vocal frame activates a few formant atoms; a percussive frame is sparse in time) and the sustained harmonic content of an accompaniment is approximately low-rank. Both properties are exactly the structural assumptions of compressed sensing and sparse coding (Donoho, 2006; Elad, 2010). Deep masking networks, in turn, can be read as unrolled sparse-recovery solvers with learned operators (Gregor & LeCun, 2010; Monga et al., 2021). Placing the classical and deep methods on one short-time Fourier transform (STFT) front end therefore lets us compare them as

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points on a structure-recovery spectrum rather than as unrelated systems.

Contributions. (i) We formalize vocal/accompaniment separation as a structured-recovery problem and instantiate four separators that share a front end and an evaluation harness (Section 5). (ii) We construct a deep sparse coding network that predicts a soft time-frequency mask by unrolling a non-negative fast proximal-gradient iteration and learning the synthesis dictionaries end-to-end, and show it is the strongest non-oracle separator in our study while being the fastest learned model at test time (Sections 5.4 and 6). (iii) We pair every method with the recovery condition that governs it, and report an honest negative result on dictionary coherence that quantifies the gap between worst-case guarantees and observed behaviour (Sections 4 and 6.1). All code, trained weights, and audio examples are released for reproducibility.

2. Background: Spectrograms, Stems, and Masking

We summarize the audio concepts the paper relies on so the development is self-contained for readers outside music processing.

From waveform to spectrogram. A recording is a pressure waveform $y(t)$ sampled at f_s samples per second. The short-time Fourier transform (STFT) slides a window of length w across the signal with hop h and takes a discrete Fourier transform in each window, producing a complex matrix $\boldsymbol{Y} \in \mathbb{C}^{F \times T}$ where $F = w/2 + 1$ frequency bins index pitch and T frames index time. Its modulus $\boldsymbol{M} = |\boldsymbol{Y}|$, the magnitude spectrogram, is a nonnegative image whose brightness at (f, t) is the energy near frequency f at time t; its argument $\angle \boldsymbol{Y}$ is the phase. The transform is invertible, so y can be reconstructed exactly from $\boldsymbol{M}$ and $\angle \boldsymbol{Y}$.

Stems and the separation goal. A finished song is a sum of source stems (vocals, drums, bass, other). We target the two-way split into vocals and accompaniment (everything else). Because the STFT is linear, the mixture spectrogram is approximately the sum of the source spectrograms, $\boldsymbol{Y} \approx \sum_c \boldsymbol{Y}_c$, and on magnitudes the sources mostly occupy different time-frequency cells, which is what makes masking viable.

Time-frequency masking. Rather than predict a source spectrogram outright, we predict a mask $\boldsymbol{M}_c \in [0, 1]^{F \times T}$ that scales each cell of the mixture and resynthesize with the mixture phase,

$$\hat{y}_c = \mathrm{STFT}^{-1}(\boldsymbol{M}_c \odot \boldsymbol{Y}). \tag{1}$$

A mask answers a local question: what fraction of the energy in cell (f, t) belongs to source c? The optimal mask under a squared-error criterion is the ideal ratio mask

$$\boldsymbol{M}_c^{\mathrm{IRM}} = \frac{|\boldsymbol{Y}_c|^2}{\sum_k |\boldsymbol{Y}_k|^2}, \tag{2}$$

which is exactly the multichannel Wiener / minimum-mean-square-error estimator when sources are modelled as locally uncorrelated Gaussians. Equation (2) requires the true sources, so it serves only as an oracle upper bound; the binary variant $\boldsymbol{M}_c^{\mathrm{IBM}} = \mathbb{1}[|\boldsymbol{Y}_c| > |\boldsymbol{Y}_{\neg c}|]$ is a second oracle. Every method we build estimates a mask without access to the sources, and no mixture-phase magnitude mask can exceed the ideal-mask ceilings.

Why sparsity and low rank. Two empirical regularities make these masks recoverable by structured optimization. A voiced sound places energy on a sparse set of harmonics (a few bright horizontal ridges per frame), so its spectrogram is approximately sparse in a harmonic-atom basis. Sustained, repeating accompaniment reuses a small set of spectral templates across many frames, so its magnitude spectrogram is approximately low-rank. These are the two structural priors compressed sensing is built to exploit, and they organize the four methods below.

3. Related Work

Factorization and low-rank models. NMF decomposes a non-negative spectrogram into non-negative spectral atoms and activations (Lee & Seung, 1999; 2000) and was applied to polyphonic audio by Smaragdis & Brown (2003). Supervised and convolutive variants (Smaragdis, 2007; Virtanen, 2007) pretrain one dictionary per source and hold it fixed at test time, which is the form we adopt. The choice of divergence matters for audio: the Itakura-Saito divergence (Févotte et al., 2009) is scale-invariant and matches a Gaussian generative model of power spectra. Low-rank methods exploit the repetitive structure of accompaniment: REPET (Rafii & Pardo, 2013) models the background as a repeating pattern, and robust principal component analysis (RPCA) separates a low-rank accompaniment from a sparse vocal (Candès et al., 2011; Huang et al., 2012). Median-filtering harmonic/percussive separation (FitzGerald, 2010) is a training-free baseline in the same family.

Dictionary learning and sparse representation. K-SVD learns overcomplete sparsifying dictionaries by alternating sparse coding with rank-one atom updates (Aharon et al., 2006; Engan et al., 1999). Sparse-representation classification (SRC) assigns a signal to the class whose sub-dictionary best reconstructs it (Wright et al., 2009); we use this rule to route time-frequency energy. Task-driven dictionary learning (Mairal et al., 2012) optimizes the dictionary for a downstream loss by differentiating through the sparse code, which is the conceptual bridge to our learned model.

Deep separation and algorithm unrolling. End-to-end networks dominate the leaderboards (Hennequin et al., 2020; Stöter et al., 2019; Défossez et al., 2021; Rouard et al., 2023; Luo & Mesgarani, 2019; Mitsufuji et al., 2022). A complementary line interprets such networks through sparse coding: LISTA unrolls the iterative shrinkage-thresholding algorithm into a fixed-depth trainable network (Gregor & LeCun, 2010); CNNs admit a multilayer convolutional-sparse-coding reading (Papyan et al., 2017); and algorithm unrolling is now a general design principle (Monga et al., 2021; Sprechmann et al., 2015). Discriminatively unfolded NMF has been used for speech separation (Le Roux et al., 2015), and supervised deep sparse coding networks have been studied for image classification (Sun et al., 2019). Our learned separator instantiates this unrolling idea for music spectrogram masking.

4. Preliminaries: Structured Recovery

We collect the recovery results that the methods rely on. Throughout, $\|\boldsymbol{x}\|_0 = |\mathrm{supp}(\boldsymbol{x})|$ counts nonzeros and $\|\boldsymbol{x}\|_1 = \sum_i |x_i|$.

Sparse recovery. Given $\boldsymbol{A} \in \mathbb{R}^{m\times N}$ with $m \ll N$ and $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ for an s-sparse $\boldsymbol{x}$, the combinatorial program $\min_{\boldsymbol{z}} \|\boldsymbol{z}\|_0$ s.t. $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}$ is NP-hard; its convex relaxation $\min_{\boldsymbol{z}} \|\boldsymbol{z}\|_1$ s.t. $\boldsymbol{A}\boldsymbol{z} = \boldsymbol{y}$ is a linear program. Exactness of the relaxation is controlled by three quantities. The spark is the size of the smallest linearly dependent column subset, and an s-sparse solution is unique iff $\mathrm{spark}(\boldsymbol{A}) > 2s$. The restricted isometry property (RIP) of order s holds with constant δ_s if

$$(1-\delta_s)\|\boldsymbol{x}\|_2^2 \le \|\boldsymbol{A}\boldsymbol{x}\|_2^2 \le (1+\delta_s)\|\boldsymbol{x}\|_2^2 \tag{3}$$

for all s-sparse $\boldsymbol{x}$; ℓ_1 recovery is exact whenever $\delta_{2s} < \sqrt{2}-1$ (Candès & Tao, 2005). The mutual coherence $\mu(\boldsymbol{A}) = \max_{i\neq j} |\langle \boldsymbol{A}_i, \boldsymbol{A}_j\rangle|$ (unit-norm columns) gives the verifiable condition

$$s < \tfrac{1}{2}\Big(1 + \tfrac{1}{\mu(\boldsymbol{A})}\Big) \tag{4}$$

under which both orthogonal matching pursuit (OMP) and ℓ_1 minimization recover every s-sparse signal (Tropp & Gilbert, 2007; Donoho, 2006). For sub-Gaussian random $\boldsymbol{A}$, RIP holds with high probability once $m \ge C\, s \log(N/s)$.

Low-rank recovery. The nuclear norm $\|\boldsymbol{X}\|_* = \sum_i \sigma_i(\boldsymbol{X})$ is the convex surrogate for rank. Robust PCA recovers a low-rank $\boldsymbol{X}$ and a sparse $\boldsymbol{E}$ from $\boldsymbol{M} = \boldsymbol{X} + \boldsymbol{E}$ by principal component pursuit,

$$\min_{\boldsymbol{X},\boldsymbol{E}} \|\boldsymbol{X}\|_* + \lambda\|\boldsymbol{E}\|_1 \quad \text{s.t.} \quad \boldsymbol{X} + \boldsymbol{E} = \boldsymbol{M}, \tag{5}$$

which succeeds with high probability at $\lambda = 1/\sqrt{\max(n_1, n_2)}$ when the low-rank term is incoherent and the support of $\boldsymbol{E}$ is random (Candès et al., 2011; Candès & Recht, 2009).

5. Method

Front end. Let $y(t)$ be a mono mixture. Its STFT is $\boldsymbol{Y} = \mathrm{STFT}(y) \in \mathbb{C}^{F\times T}$ with magnitude $\boldsymbol{M} = |\boldsymbol{Y}|$ and phase $\angle\boldsymbol{Y}$. Every separator produces a soft mask $\boldsymbol{M}_c \in [0,1]^{F\times T}$ per source c, and resynthesizes $\hat{y}_c = \mathrm{STFT}^{-1}(\boldsymbol{M}_c \odot \boldsymbol{Y})$ using the mixture phase. Given non-negative source magnitude estimates $\widehat{\boldsymbol{M}}_c$, we use Wiener-style masks $\boldsymbol{M}_c = \widehat{\boldsymbol{M}}_c^2 / \sum_k \widehat{\boldsymbol{M}}_k^2$.

5.1. Low-rank plus sparse (RPCA)

We solve Equation (5) on the magnitude spectrogram by the inexact augmented Lagrange multiplier method. With the soft-threshold $\mathrm{soft}_\tau(x) = \mathrm{sgn}(x)\max(|x| - \tau, 0)$ and the singular-value threshold $\mathrm{SVT}_\tau(\boldsymbol{Z}) = \boldsymbol{U}\,\mathrm{soft}_\tau(\boldsymbol{\Sigma})\boldsymbol{V}^\top$ for $\boldsymbol{Z} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, the iteration is

$$\boldsymbol{X} \leftarrow \mathrm{SVT}_{1/\mu}\big(\boldsymbol{M} - \boldsymbol{E} + \tfrac{1}{\mu}\boldsymbol{\Lambda}\big), \tag{6}$$

$$\boldsymbol{E} \leftarrow \mathrm{soft}_{\lambda/\mu}\big(\boldsymbol{M} - \boldsymbol{X} + \tfrac{1}{\mu}\boldsymbol{\Lambda}\big), \tag{7}$$

$$\boldsymbol{\Lambda} \leftarrow \boldsymbol{\Lambda} + \mu(\boldsymbol{M} - \boldsymbol{X} - \boldsymbol{E}), \tag{8}$$

with μ increased geometrically. The low-rank $\boldsymbol{X}$ is the accompaniment estimate and the sparse $\boldsymbol{E}$ the vocal proxy; this is principal component pursuit applied to singing-voice separation (Huang et al., 2012). The method is training-free.

5.2. Supervised NMF

We pre-train one non-negative dictionary per source on isolated stems by minimizing $\|\boldsymbol{M}_c - \boldsymbol{D}_c\boldsymbol{H}_c\|$ subject to $\boldsymbol{D}_c, \boldsymbol{H}_c \ge 0$. With the Kullback-Leibler divergence (the natural loss for magnitude spectra (Févotte et al., 2009)), the Lee-Seung multiplicative updates (Lee &

Seung, 2000) are

$$\boldsymbol{H} \leftarrow \boldsymbol{H} \odot \frac{\boldsymbol{D}^\top(\boldsymbol{M} \oslash \boldsymbol{D}\boldsymbol{H})}{\boldsymbol{D}^\top\boldsymbol{1}}, \qquad \boldsymbol{D} \leftarrow \boldsymbol{D} \odot \frac{(\boldsymbol{M} \oslash \boldsymbol{D}\boldsymbol{H})\boldsymbol{H}^\top}{\boldsymbol{1}\boldsymbol{H}^\top}, \tag{9}$$

where $\odot, \oslash$ are elementwise. These updates are not heuristic: they are a majorization-minimization scheme. For the divergence $D(\boldsymbol{M} \,\|\, \boldsymbol{D}\boldsymbol{H})$ one builds an auxiliary function $G(\boldsymbol{H}, \boldsymbol{H}') \geq D(\boldsymbol{M} \,\|\, \boldsymbol{D}\boldsymbol{H})$ with equality at $\boldsymbol{H}' = \boldsymbol{H}$, using the convexity of $-\log$ and Jensen's inequality on the latent assignment $\boldsymbol{D}_{ik}H_{kj}/(\boldsymbol{D}\boldsymbol{H})_{ij}$; minimizing $G(\cdot, \boldsymbol{H}')$ in closed form yields the multiplicative step, so each update cannot increase the loss and the iterates converge to a stationary point. The factors stay nonnegative automatically because every quantity in the ratio is nonnegative. At test time we fix $\boldsymbol{D} = [\boldsymbol{D}_v \mid \boldsymbol{D}_a]$ and solve only for activations $\boldsymbol{H}$, then reconstruct $\widehat{\boldsymbol{M}}_v = \boldsymbol{D}_v\boldsymbol{H}_v$ and $\widehat{\boldsymbol{M}}_a = \boldsymbol{D}_a\boldsymbol{H}_a$. This is the supervised NMF template of Smaragdis (2007); the Kullback-Leibler choice corresponds to a Poisson observation model on spectrogram counts, while the Itakura-Saito divergence ($\beta = 0$) corresponds to a multiplicative-Gaussian model of power spectra (Févotte et al., 2009).

5.3. Discriminative dictionaries with SRC

We replace each NMF dictionary with an overcomplete K-SVD dictionary trained to sparsely code spectrogram frames, $\min_{\boldsymbol{D},\boldsymbol{X}} \|\boldsymbol{Y} - \boldsymbol{D}\boldsymbol{X}\|_F^2$ s.t. $\|\boldsymbol{x}_i\|_0 \leq s$. K-SVD (Aharon et al., 2006) alternates OMP sparse coding with a per-atom update: writing $\boldsymbol{D}\boldsymbol{X} = \sum_j \boldsymbol{D}_j \boldsymbol{x}^{j\top}$ and forming the residual $\boldsymbol{E}_j = \boldsymbol{Y} - \sum_{k \neq j} \boldsymbol{D}_k \boldsymbol{x}^{k\top}$ restricted to the atom's active set, the rank-one SVD $\boldsymbol{E}_j^R = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ updates $\boldsymbol{D}_j \leftarrow \boldsymbol{u}_1$ and the coefficients $\leftarrow \sigma_1\boldsymbol{v}_1$. The rank-one update is optimal in the Eckart-Young sense: $(\boldsymbol{u}_1, \sigma_1\boldsymbol{v}_1)$ is the exact minimizer of $\|\boldsymbol{E}_j^R - \boldsymbol{D}_j\boldsymbol{x}^{j\top}\|_F$ over all unit atoms and coefficient rows, so the atom update is a coordinate descent step that cannot increase the objective, and restricting to the active set $\{i : \boldsymbol{x}_i^j \neq 0\}$ preserves the sparsity pattern set by the coding step. At test time we sparse-code each mixture frame on $[\boldsymbol{D}_v \mid \boldsymbol{D}_a]$ by OMP and route energy by the SRC residual rule (Wright et al., 2009): $c^\star = \arg\min_c \|\boldsymbol{y} - \boldsymbol{D}_c\boldsymbol{\alpha}_c\|_2$. The coherence condition Equation (4) governs when this per-frame code is unique, linking the separator's reliability to a measurable property of the learned dictionary (Section 6.1).

5.4. Deep sparse coding network

Our learned separator predicts the vocal mask by unrolling a non-negative sparse solver and learning its dictionaries end-to-end. For an input frame $\boldsymbol{x} \in \mathbb{R}^F$ a composite module computes two stacked non-negative elastic-net codes,

$$\boldsymbol{\alpha}^{(1)} = \arg\min_{\boldsymbol{\alpha} \geq 0} \tfrac{1}{2}\|\boldsymbol{x} - \boldsymbol{D}^{(1)}\boldsymbol{\alpha}\|_2^2 + \lambda_1\|\boldsymbol{\alpha}\|_1 + \tfrac{\lambda_2}{2}\|\boldsymbol{\alpha}\|_2^2, \tag{10}$$

$$\boldsymbol{\alpha}^{(2)} = \arg\min_{\boldsymbol{\alpha} \geq 0} \tfrac{1}{2}\|\boldsymbol{\alpha}^{(1)} - \boldsymbol{D}^{(2)}\boldsymbol{\alpha}\|_2^2 + \lambda_1\|\boldsymbol{\alpha}\|_1 + \tfrac{\lambda_2}{2}\|\boldsymbol{\alpha}\|_2^2, \tag{11}$$

with $\boldsymbol{D}^{(1)} \in \mathbb{R}^{F \times n_u}$ overcomplete ($n_u > F$) and $\boldsymbol{D}^{(2)} \in \mathbb{R}^{n_u \times n_d}$ undercomplete ($n_d < n_u$). Each subproblem is solved by K steps of a non-negative fast proximal-gradient iteration: with step size $1/L$, L the squared spectral norm of the dictionary (obtained by power iteration, so no on-device SVD is needed),

$$\boldsymbol{\alpha}_k = \left[\boldsymbol{z}_k - \tfrac{1}{L}\boldsymbol{D}^\top(\boldsymbol{D}\boldsymbol{z}_k - \boldsymbol{x}) - \tfrac{\lambda_1}{L}\right]_+, \tag{12}$$

followed by the Nesterov extrapolation $t_{k+1} = \frac{1}{2}(1 + \sqrt{1 + 4t_k^2})$, $\boldsymbol{z}_{k+1} = \boldsymbol{\alpha}_k + \frac{t_k - 1}{t_{k+1}}(\boldsymbol{\alpha}_k - \boldsymbol{\alpha}_{k-1})$. Every operation is a matrix multiply or a clamp, so the unrolled solver is differentiable and runs on a GPU. Stacking L_m composite modules and a linear head with a sigmoid yields a mask $\boldsymbol{M}_v \in [0,1]^{F \times T}$. The dictionaries $\boldsymbol{D}^{(\ell)}$ and shrinkage parameters λ_1, λ_2 are trained by backpropagation through the unrolled iterations, minimizing a masked-magnitude reconstruction loss against the ideal ratio mask with an explicit sparsity penalty on the codes,

$$\mathcal{L} = \|(\boldsymbol{M}_v - \boldsymbol{M}_v^{\mathrm{IRM}}) \odot \boldsymbol{M}\|_1 + \beta \sum_\ell \|\boldsymbol{\alpha}^{(\ell)}\|_1. \tag{13}$$

Algorithm 1 Composite module forward pass (one of L_m). All steps are matrix multiplies and clamps, so the pass is differentiable and runs on a GPU.

1: Input: frame $\boldsymbol{x}$, dictionaries $\boldsymbol{D}^{(1)}, \boldsymbol{D}^{(2)}$, shrinkage λ_1, λ_2, steps K
2: $\boldsymbol{D} \leftarrow \boldsymbol{D}^{(1)}$; $L \leftarrow \sigma_{\max}(\boldsymbol{D})^2 + \lambda_2$ (power iteration)
3: $\boldsymbol{\alpha} \leftarrow \boldsymbol{0}$; $\boldsymbol{z} \leftarrow \boldsymbol{0}$; $t \leftarrow 1$
4: for $k = 1$ to K do
5: $\quad \boldsymbol{g} \leftarrow \boldsymbol{D}^\top(\boldsymbol{D}\boldsymbol{z} - \boldsymbol{x}) + \lambda_2\boldsymbol{z}$ (gradient)
6: $\quad \boldsymbol{\alpha}^+ \leftarrow \left[\boldsymbol{z} - \boldsymbol{g}/L - \lambda_1/L\right]_+$ (non-neg. shrinkage)
7: $\quad t^+ \leftarrow \frac{1}{2}(1 + \sqrt{1 + 4t^2})$
8: $\quad \boldsymbol{z} \leftarrow \boldsymbol{\alpha}^+ + \frac{t-1}{t^+}(\boldsymbol{\alpha}^+ - \boldsymbol{\alpha})$; $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha}^+$; $t \leftarrow t^+$
9: end for
10: repeat lines 2–9 with $\boldsymbol{D}^{(2)}$ on input $\boldsymbol{\alpha}$ to obtain $\boldsymbol{\alpha}^{(2)}$
11: return $\boldsymbol{\alpha}^{(2)}$

Algorithm 1 gives the per-module computation. Equations (10) to (12) make explicit that the network is a learned sparse-recovery operator: Equation (12) is one unrolled iteration of the same proximal solver used for ℓ_1 recovery, with $\boldsymbol{D}$ promoted to a trainable parameter, exactly the LISTA construction (Gregor & LeCun, 2010) specialized to non-negative codes and a masking output. The clamp $[\cdot]_+$ is the proximal operator of the indicator of the nonnegative orthant composed with the soft-threshold, i.e. the prox of $\lambda_1\|\boldsymbol{\alpha}\|_1 + \iota_{\boldsymbol{\alpha}\geq 0}(\boldsymbol{\alpha})$, so each layer is a genuine proximal-gradient step. Run to convergence rather than truncated, the accelerated iteration attains the optimal first-order rate $F(\boldsymbol{\alpha}_k) - F(\boldsymbol{\alpha}^\star) = O(1/k^2)$ (Beck & Teboulle, 2009); unrolling a fixed K trades this asymptotic guarantee for a learned operator that reaches comparable quality in a handful of layers, which is why inference is a single fast forward pass rather than an iterative solve (Section 6).

5.5. Joint sparsity for stereo

Stereo channels share active sources, so their codes share a row support. Stacking the per-frame channel magnitudes into $\boldsymbol{Y} = [\boldsymbol{y}_L\ \boldsymbol{y}_R] \in \mathbb{R}^{F\times 2}$, we apply simultaneous OMP (Tropp et al., 2006), which scores atoms by the row norm $\|(\boldsymbol{D}^\top \boldsymbol{R})_{n,:}\|_2$ of the residual and so forces both channels to agree on the active atoms. The mixed norm $\|\boldsymbol{X}\|_{1,2} = \sum_n \|\boldsymbol{X}_{n,:}\|_2$ is the convex surrogate, and the shared support strictly improves identifiability: a row-s-sparse $\boldsymbol{X}$ is unique once $s < (\mathrm{spark}(\boldsymbol{A}) - 1 + \mathrm{rank}(\boldsymbol{Y}))/2$.

6. Experiments

Data and protocol. We use the MUSDB18 benchmark (Rafii et al., 2017), separating each mixture into vocals and accompaniment. Dictionaries and the deep model are trained on the training split; all numbers are reported on 25 held-out test tracks at 22.05 kHz with a 2048-point STFT and hop 512. We report the median scale-invariant SDR (Le Roux et al., 2019) and SDR (Vincent et al., 2006) per source. Two oracle masks bound any masking method: the ideal ratio mask (IRM) and ideal binary mask (IBM).

Methods. We compare two training-free baselines, median-filtering harmonic/percussive separation (HPSS) (FitzGerald, 2010) and REPET-SIM (Rafii & Pardo, 2013), against the four structured-recovery separators of Section 5: RPCA, supervised NMF (K=64 atoms per source, KL divergence), discriminative K-SVD (K=96, sparsity s=10), and the deep sparse coding network (SCN; three composite modules, n_u=192, n_d=96, K=18 unrolled steps, 0.32M parameters, trained on Apple MPS).

Table 1. Median SI-SDR and SDR (dB) on 25 MUSDB18 test tracks, with mean inference time per 6 s clip. Trivial floors, training-free baselines, structured-recovery methods, and oracle ceilings. Best non-oracle in bold.

	SI-SDR		SDR	Time
Method	Vocals	Accomp.	Vocals	(s)
Trivial floors				
Mixture (no split)	−3.7	4.0	—	0.00
Random mask	−3.9	2.9	—	0.01
Low-rank (rank 4)	−0.1	6.1	—	0.06
Training-free baselines				
HPSS (median)	−11.9	−1.4	0.3	0.25
REPET-SIM	−0.7	2.9	2.5	0.08
Structured-recovery methods				
RPCA	−3.1	0.2	−1.3	1.14
NMF (supervised)	0.5	5.1	1.8	0.46
K-SVD + SRC	0.8	7.1	2.7	0.28
SCN (ours)	**1.6**	**7.7**	**3.8**	0.08
Mask fusion	1.4	**7.8**	—	0.82
Oracle ceilings				
Oracle IRM	10.1	15.3	10.2	—
Oracle IBM	10.2	15.8	10.5	—

Main results. Table 1 and Figure 3 show a consistent ordering. The trivial floors calibrate the scale: returning the mixture unchanged already scores +4.0 dB on accompaniment, because the accompaniment dominates the mixture energy, so accompaniment SI-SDR must be read against this floor rather than against zero; a random mask is near the no-information floor on both sources. The training-free baselines are mixed: HPSS over-suppresses tonal vocal energy (its median vocal SI-SDR falls below even the mixture floor), and REPET-SIM helps only when the background strictly repeats. The supervised dictionary methods improve sharply over all floors, and the deep sparse coding network is the best non-oracle separator on both sources (1.6 dB vocals, 7.7 dB accompaniment) while also being the fastest learned model, because its forward pass is a fixed number of matrix multiplies rather than an iterative solve. A simple geometric-mean fusion of the three learned masks edges out the best single model on accompaniment (7.8 dB), indicating the methods make partly complementary errors. All methods remain below the IRM/IBM oracle ceilings, as any mixture-phase magnitude mask must.

Per-track variance. Medians hide spread. Figure 1 plots the per-track vocal SI-SDR distribution. The structured-recovery methods are not uniformly better track by track: their boxes overlap, and every method has a long lower tail on dense, vocal-quiet mixtures where the sparsity prior is weakest. The deep model

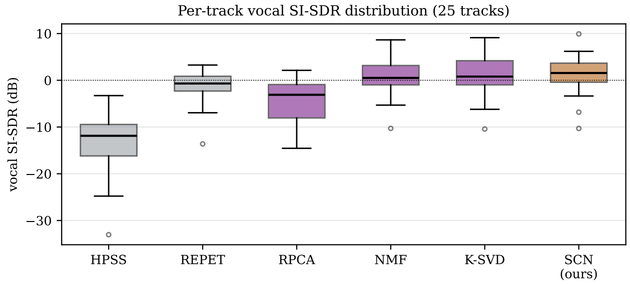

Figure 1. Per-track vocal SI-SDR over the 25 test tracks. Boxes span the interquartile range; the deep model shifts the full distribution, not only the median.

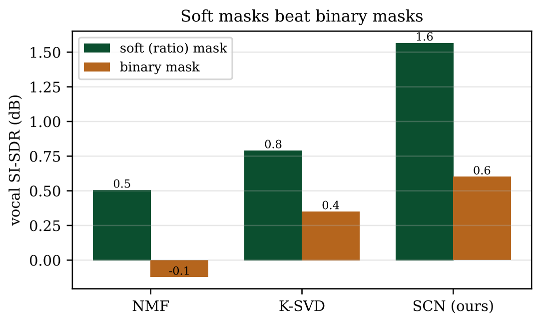

Figure 2. Soft (ratio) masks beat hard binary masks for every learned method, by up to 1 dB of vocal SI-SDR.

has the highest median and upper quartile, and shifts the whole distribution rightward relative to the classical methods rather than only improving easy tracks.

Soft versus binary masks. The masking strategy matters independently of the model. Figure 2 compares each learned method's soft (ratio) mask against its hard-thresholded binary version. Soft masks are uniformly better, by roughly 1 dB for the deep model (1.6 vs 0.6 dB), because a continuous gain preserves the partial-overlap cells that a binary decision discards. This mirrors the gap between the IRM and IBM oracles and motivates our soft-mask training target.

Stereo joint sparsity. Evaluating the simultaneous-OMP stereo separator of Section 5.5 on 10 stereo test tracks gives median SI-SDR 0.9 dB vocals and 7.3 dB accompaniment, matching the mono K-SVD dictionary it is built on while producing a coherent stereo image, since the shared row support forces the two channels to agree on active atoms. This confirms the joint-sparsity extension is sound; a larger stereo study is left to future work.

Spectrogram structure. Figure 4 shows the sparsity the methods exploit: the vocal magnitude concentrates on a thin harmonic scaffold while the accompaniment forms a denser, lower-rank stack. Figure 5 reports the RPCA decomposition and confirms the low-rank prior: the magnitude spectrogram's singular spectrum decays sharply and the proximal iteration converges to a relative residual below 10^{-7}.

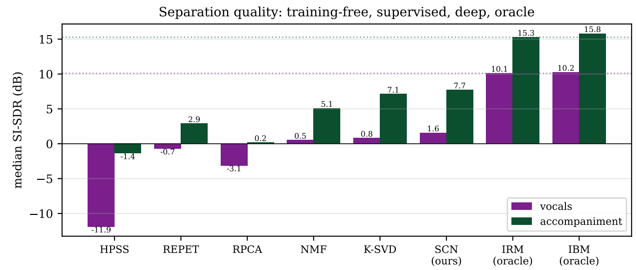

Figure 3. Median SI-SDR by method. Dotted lines mark the oracle ceilings. Supervision and unrolled learning each add a clear increment over the training-free baselines.

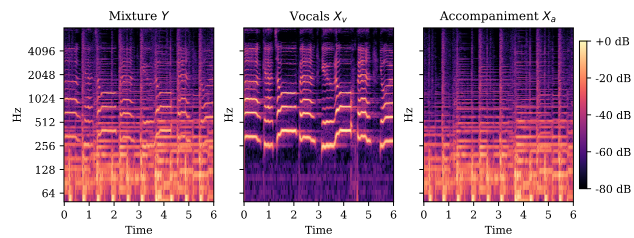

Figure 4. Log-magnitude STFT of a test clip. Vocals are sparse on a formant scaffold; accompaniment is denser and lower-rank.

Ablations. Table 2 sweeps the central hyperparameters and the trends are informative rather than uniform. For supervised NMF the vocal SI-SDR is highest at the smallest dictionary (K=16) and declines as K grows: with limited training data a small dictionary acts as a stronger prior and generalizes better, a regularization effect rather than a capacity limit. K-SVD separation improves with the per-frame sparsity budget up to s=10 and then saturates, so richer codes help up to the point where added atoms stop explaining new source-specific structure. For RPCA the vocal SI-SDR increases monotonically with the penalty λ: a larger penalty pushes more energy into the low-rank accompaniment and leaves a cleaner sparse vocal residual, so the canonical $\lambda = 1/\sqrt{\max(F,T)}$ is conservative for the singing-voice task even though it is the value certified for exact low-rank/sparse recovery.

Learned dictionaries. Figure 6 shows the supervised NMF atoms sorted by spectral centroid: vocal atoms concentrate harmonic ridges in the formant region

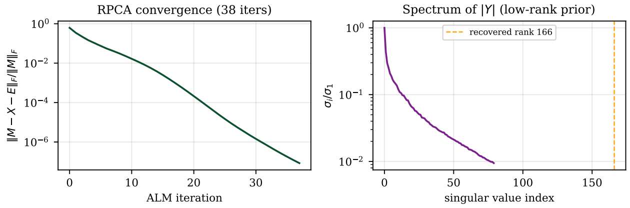

Figure 5. Left: relative residual of the principal-component-pursuit iteration. Right: normalized singular spectrum of the mixture magnitude, justifying the low-rank accompaniment prior.

Table 2. Ablations (median vocal SI-SDR, dB, 12 tracks).

NMF atoms K	16	32	64	128
Vocal SI-SDR	2.2	1.9	1.8	1.5
K-SVD sparsity s	3	5	10	15
Vocal SI-SDR	1.4	2.3	**2.7**	2.7
RPCA λ scale	0.5	1.0	1.5	2.0
Vocal SI-SDR	−3.9	−3.6	−3.2	**−2.3**

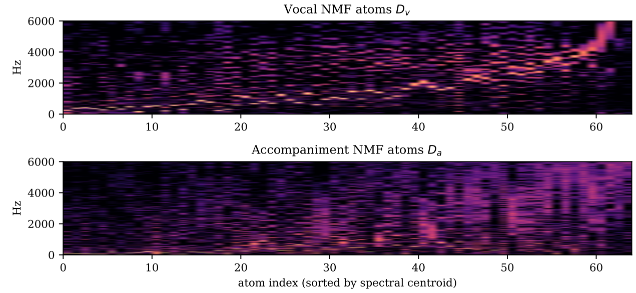

Figure 6. Supervised NMF dictionaries, atoms sorted by spectral centroid. Vocal atoms (top) versus accompaniment atoms (bottom).

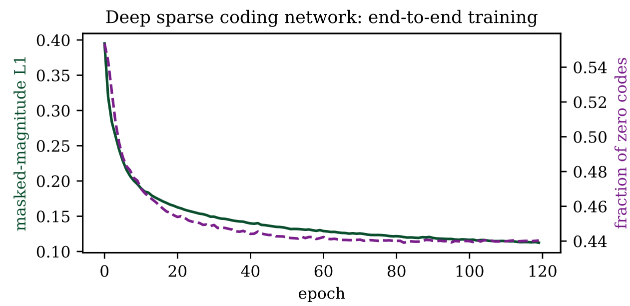

Figure 7. Deep sparse coding network training. Masked-magnitude reconstruction loss (left axis) and fraction of zero codes (right axis) across epochs.

while accompaniment atoms spread lower and wider, the parts-based structure NMF is designed to recover. Figure 7 shows that the deep model trains stably and keeps its intermediate codes sparse, confirming the sparsity prior is active rather than decorative.

6.1. Coherence: a negative result

The recovery condition Equation (4) requires low coherence, but learned overcomplete dictionaries are built for reconstruction, not incoherence. Figure 8 reports the off-diagonal Gram histogram of the K-SVD vocal dictionary: the coherence is $\mu \approx 0.85$, so Equation (4) certifies exact recovery only for $s < \frac{1}{2}(1 + 1/\mu) \approx 1.1$, far below the $s{=}10$ at which the method operates well. The worst-case guarantee is therefore vacuous in this regime, yet separation succeeds. This is the expected average-case-versus-worst-case gap: coherence bounds the worst sparse vector, while typical music frames are far from that adversarial case. We report it because it sets honest expectations for how much of the classical theory transfers to trained audio dictionaries.

7. Discussion

The results are deliberately scoped: short clips, modest dictionaries, and a 0.32M-parameter network trained briefly on a laptop GPU. Large end-to-end systems reach substantially higher vocal SI-SDR with three to four orders of magnitude more parameters and data (Rouard et al., 2023; Mitsufuji et al., 2022). The value here is the unifying view and the controlled comparison: on one front end, the same axis of increasing structure-learning, from training-free decomposition, to supervised factorization, to discriminative dictionaries, to an unrolled deep solver, produces a monotone improvement, and the unrolled solver is both the most accurate and the cheapest to run at inference. Two limitations are intrinsic to the present setup. First, magnitude masking with mixture phase upper-bounds every method at the IRM oracle; a complex-mask or phase-reconstruction extension would lift the ceiling (Luo & Mesgarani, 2019). Second, the deep model is trained per frame and ignores temporal context; convolutional or recurrent composite modules are a natural next step. The stereo joint-sparsity extension of Section 5.5 also remains to be evaluated at scale.

8. Conclusion

We presented a unified compressed-sensing view of music source separation, casting low-rank decomposition, supervised NMF, discriminative dictionary learning, and a deep unrolled sparse coder as points on one structure-recovery spectrum over the STFT. On

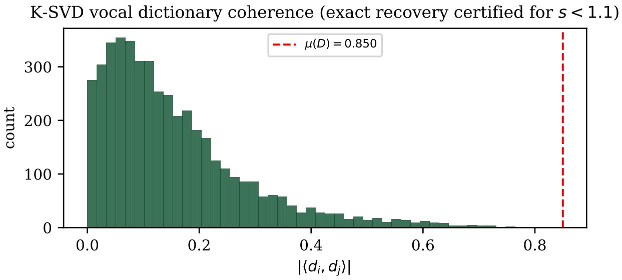

Figure 8. Off-diagonal coherence of the learned K-SVD vocal dictionary. The worst-case recovery bound certifies only $s \lesssim 1$, but separation at s=10 works, evidence of an average-case-versus-worst-case gap.

MUSDB18 the methods order cleanly, the unrolled deep sparse coding network is the strongest non-oracle separator while being the fastest learned model, and a coherence analysis delimits where the classical recovery theory does and does not transfer to learned audio dictionaries. Code, models, and audio are released.

Reproducibility

All code, trained dictionaries and network weights, the evaluation harness, and curated before/after audio are available at https://github.com/takakhoo/ENGS109_Final_Project.

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